Tutorial 6: An Introduction to Probability

10 November 2025

1. A game is played by rolling two dice: a tetrahedral one with vertices labeled 1 through 4, and a cubic one with faces labeled 1 through 6.

   (a) Describe the sample space of possible outcomes when both dice are rolled at the same time.

   (b) A player's score is computed by taking the positive difference between the numbers rolled on both dice.

      i. Find the probability that a player obtains a score of exactly zero.

      ii. Find the probability that a player obtains a score greater than or equal to three.

   (c) The game is played 150 times. Find the expected number of times that a prime number score will occur.

(a) We use a table to list out all the possible outcomes.

	1	2	3	4
1	$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$
2	$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$
3	$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$
4	$(4,1)$	$(4,2)$	$(4,3)$	$(4,4)$
5	$(5,1)$	$(5,2)$	$(5,3)$	$(5,4)$
6	$(6,1)$	$(6,2)$	$(6,3)$	$(6,4)$

	1	2	3	4
1	0	1	2	3
2	1	0	1	2
3	2	1	0	1
4	3	2	1	0
5	4	3	2	1
6	5	4	3	2

(b) We can modify the table to instead contain the differences between each pair of values, as shown above. Let X be the positive difference between the two dice rolls.

   i. A player obtains a zero in 4 of the 24 possible outcomes. Thus,

$$P(X = 0) = \frac{4}{24} = \frac{1}{6}.$$

   ii. A score greater than equal to three occurs in 7 out of the 24 possible outcomes. Hence,

$$P(X \geq 3) = \frac{7}{24}.$$

(c) A prime number occurs when X is either $2, 3$, or 5. Thus,

$$\begin{aligned} P(X \text{ is prime}) &= P(X = 2) + P(X = 3) + P(X = 5) \\ &= \frac{6}{24} + \frac{4}{24} + \frac{1}{24} = \frac{11}{24}. \end{aligned}$$

Let Y be the number of times the positive difference between the two rolls is a prime number when the game is played 150 times. Then, $Y \sim \text{Binom}(150, 11/24)$. The expected value of Y is thus

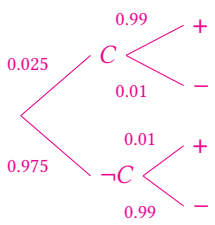
$$\mathbb{E}[Y] = 150 \cdot \frac{11}{24} = \frac{275}{4}.$$

2. A COVID-19 Antigen Rapid Test claims to have a 99% accuracy rate: a person with COVID-19 will test positive 99% of the time, while a person without COVID-19 will test negative 99% of the time. Recent medical surveys suggest that about 2.5% of Singapore's population currently have COVID-19.

   Suppose you take the test and it comes back positive. Determine the probability that you actually have COVID-19.

   Let C be the event that you have COVID-19. We may assume that $P(C) = 0.025$, such that $P(\neg C) = 1 - 0.025 = 0.975$.

   We construct a tree diagram.



$$P(C \cap +) = 0.025 \cdot 0.099 = 0.02475$$
$$P(C \cap -) = 0.025 \cdot 0.01 = 0.00025$$
$$P(\neg C \cap +) = 0.975 \cdot 0.01 = 0.00975$$
$$P(\neg C \cap -) = 0.975 * 0.99 = 0.96525$$

   We now compute the probability that you have COVID-19, given the condition that you test positive:

$$\begin{aligned} P(C \mid +) &= \frac{P(C \cap +)}{P(+)} = \frac{P(C \cap +)}{P(C \cap +) + P(\neg C \cap +)} \\ &= \frac{0.02475}{0.02475 + 0.00975} \approx 0.717391\ldots \end{aligned}$$

   Thus, there is about a 72% chance that you have COVID-19.

3. Let X be a random variable. Prove that

$$\operatorname{Var}(X) = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2.$$

   Proof. Let $\mu = \mathbb{E}[X]$. Then,

$$\begin{aligned} \operatorname{Var}(X) &= \mathbb{E}\left[(X-\mu)^2\right] = \mathbb{E}\left[X^2 - 2\mu X + \mu^2\right] = \mathbb{E}\left[X^2\right] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}\left[X^2\right] - 2\mu^2 + \mu^2 = \mathbb{E}\left[X^2\right] - \mu^2 = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2. \end{aligned}$$

   □

4. You play a game in which a fair coin is tossed until a tails appears for the first time. If the tails appears on the nth flip, you win $\$2^n$. What is your expected earnings for playing the game? Would you be willing to pay $\$1,000,000$ to play this game once?

   Let Y be the number of coin flips it takes until the first tail appears. Then, $Y \sim \operatorname{Geom}(1/2)$, and so

$$P(Y = n) = \left(\frac{1}{2}\right)^{n-1}\left(\frac{1}{2}\right) = \frac{1}{2^n}, \quad n = 1, 2, 3, \ldots.$$

   The player's winnings is given by $X = 2^Y$. Hence,

$$\mathbb{E}[X] = \mathbb{E}\left[2^Y\right] = \sum_{n=1}^{\infty} 2^n \cdot \frac{1}{2^n} = \sum_{n=1}^{\infty} 1 \to \infty.$$

5. It is observed that a certain lecturer makes, on average, two typographical errors every 10000 words. If each page in a set of lecture notes consists of 700 words, what is the probability that the page contains at least one typographical error?

   Let X be the number of words with errors on a page of lecture notes. Then, $X \sim \text{Binom}(700, 2/10000)$, with parameters $n = 700$ (total number of words) and $p = 2/10000$ (probability of a typographical error). Then, the probability that there is more than more than one error on the page is

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \binom{700}{0}\left(\frac{2}{10000}\right)^0\left(\frac{9998}{10000}\right)^{700} \approx 0.13065\ldots.$$

   Since n is large and p is small, we may view $X \sim \text{Pois}(\lambda)$ as Poisson distributed with mean

$$\lambda = np = 700 \cdot \frac{2}{10000} = 0.14.$$

   Then,

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-0.14}\frac{0.14^0}{0!} \approx 0.13064\ldots.$$

6. Let X be a discrete random variable. The *hazard function* of X is defined as

$$h_X(x) = P(X = x + 1 \mid X > x).$$

   This computes the probability that an event occurs at a specific time, given that it has not happened yet. Show that if $X \sim \text{Geom}(p)$ is geometrically distributed with probability of success p, then its hazard function must be constant. This shows that the geometric distribution is *memoryless*: that is, the distribution "forgets" that we have already waited time x when waiting for another time $x + 1$.

   Let $X \sim \text{Geom}(p)$. We may express the hazard function as

$$P(X = x + 1 \mid X > x) = \frac{P(X = x + 1 \cap X > x)}{P(X > x)} = \frac{P(X = x + 1)}{P(X > x)}.$$

   To compute the denominator, observe that

$$P(X > x) = 1 - P(X \leq x) = 1 - \sum_{k=1}^{x}(1-p)^{k-1}p.$$

   The sum is a geometric series with first term p and common ratio $1 - p$:

$$\sum_{k=1}^{x}(1-p)^{k-1}p = p \cdot \frac{1-(1-p)^x}{1-(1-p)} = 1 - (1-p)^x.$$

   In particular, we have

$$P(X > x) = 1 - (1 - (1-p)^x) = (1-p)^x.$$

   This is precisely the probability that the first x trials fail. Hence, the hazard function is given by

$$h_X(x) = \frac{P(X = x + 1)}{P(X > x)} = \frac{(1-p)^x p}{(1-p)^x} = p,$$

   which is constant.

7. Let X be a random variable with probability density function

$$p(x) = \begin{cases} c\left(1-x^2\right) & -1 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Determine the value of c.

(b) Deteremine the cumulative distribution function of X.

(c) Determine $\mathbb{E}[X]$ and $\operatorname{Var}(X)$.

(a) Since $p(x)$ is a probability density function, we have

$$\int_{\mathbb{R}} p(x)\ dx = 1.$$

We compute the integral:

$$\begin{aligned} \int_{\mathbb{R}} p(x)\ dx &= \int_{-\infty}^{-1} 0\, dx + \int_{-1}^{1} c\left(1-x^2\right)\ dx + \int_{1}^{\infty} 0\, dx \\ &= 0 + c\left[x - \frac{x^3}{3}\right]_{-1}^{1} + 0 = c\left[\left(1-\frac{1}{3}\right) - \left(-1+\frac{1}{3}\right)\right] = \frac{4}{3}c. \end{aligned}$$

We thus require that

$$\frac{4}{3}c = 1 \quad \Longrightarrow \quad c = \frac{3}{4}.$$

(b) The cumulative distribution function is given by

$$F(t) = \int_{-\infty}^{t} p(x)\ dx.$$

Thus, when $t < -1$, $F(t) = 0$. When $-1 \leq t \leq 1$,

$$\begin{aligned} F(t) &= \int_{\infty}^{t} p(x)\ dx = \int_{-1}^{t} \frac{3}{4}\left(1-x^2\right) = \frac{3}{4}\left[x - \frac{x^3}{3}\right]_{-1}^{t} \\ &= \frac{3}{4}\left[\left(t - \frac{t^3}{3}\right) - \left(-1+\frac{1}{3}\right)\right] = \frac{1}{4}\left[3t - t^3 + 2\right]. \end{aligned}$$

When $t > 1$, we have $F(t) = 1$.

(c) The expected value is given by

$$\begin{aligned} \mathbb{E}[X] &= \int_{\mathbb{R}} x \cdot p(x)\ dx = \int_{-1}^{1} x \cdot \frac{3}{4}\left(1-x^2\right)\ dx \\ &= \frac{3}{4}\left[\frac{x^2}{2} - \frac{x^4}{4}\right]_{-1}^{1} = \frac{3}{4}\left(\left(\frac{1}{2}-\frac{1}{4}\right) - \left(\frac{1}{2}-\frac{1}{4}\right)\right) = 0. \end{aligned}$$

Since $\mathbb{E}[X] = 0$, the variance is given by

$$\begin{aligned} \operatorname{Var}(X) &= \int_{\mathbb{R}} (x - \mathbb{E}[X])^2 \cdot p(x)\ dx = \int_{-1}^{1} (x-0)^2 \cdot \frac{3}{4}\left(1-x^2\right)\ dx \\ &= \frac{3}{4}\left[\frac{x^3}{3} - \frac{x^5}{5}\right]_{-1}^{1} = \frac{3}{4}\left(\left(\frac{1}{3}-\frac{1}{5}\right) - \left(-\frac{1}{3}+\frac{1}{5}\right)\right) = \frac{1}{5}. \end{aligned}$$

8. Let T be an exponentially distributed random variable, modeling the waiting time for a Poisson event. Show that

$$P(T > s+t \mid T > s) = P(T > t).$$

This shows that the exponential distribution is *memoryless*: it "forgets" that we have already waited time s when waiting for another time t.

Suppose that $T \sim \text{Exp}(\lambda)$. Then, for $t \geq 0$,

$$\begin{aligned} P(T > s+t \mid T > s) &= \frac{P(T > s+t \cap T > s)}{P(T > s)} = \frac{P(T > s+t)}{P(T > s)} \\ &= \frac{1 - P(T \leq s+t)}{1 - P(T \leq s)} = \frac{1 - \left(1 - e^{-\lambda(s+t)}\right)}{1 - \left(1 - e^{-\lambda s}\right)} \\ &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(T > t). \end{aligned}$$

In particular, $P(T > t) = e^{-\lambda t}$ is independent of s.

9. The time T, in days, it takes for a hiker lost in the woods to be rescued is a continuous random variable with probability density function

$$f(t) = \frac{1}{1.5} e^{-t/1.5}, \quad t > 0.$$

(a) Determine the cumulative distribution function of T.

(b) Determine the expected time it takes to rescue a hiker.

(c) Determine the variance of T.

(d) If a hiker has been lost for three days, what is the probability that he will be found the next day?

Observe that $T \sim \text{Exp}(1/1.5)$.

(a) We compute the cumulative distribution function:

$$\begin{aligned} F(t) &= \int_{-\infty}^{t} f(x)\, dx = \int_0^t \frac{1}{1.5} e^{-x/1.5}\, dx \\ &= \frac{1}{1.5}\left[-1.5 e^{-x/1.5}\right]_0^t = -\left(e^{-t/1.5} - e^0\right) \\ &= 1 - e^{-t/1.5}. \end{aligned}$$

(b) Since T is exponentially distributed, we have

$$\mathbb{E}[T] = \frac{1}{\lambda} = 1.5.$$

Thus, the expected time it takes to rescue a hiker is 1.5 days.

(c) The variance of the exponential distribution is given by

$$\text{Var}(T) = \frac{1}{\lambda^2} = 1.5^2 = 2.25.$$

(d) Since the exponential distribution is memoryless,

$$\begin{aligned} P(\text{found the next day} \mid \text{lost for three days}) &= P(\text{found the next day}) \\ &= P(T < 1) = F(1) \\ &= 1 - e^{-1/1.5} \approx 0.48658\ldots. \end{aligned}$$

10. A small machine is constructed with five identical parts that function independently of one another. In order for the machine to work, any three of the five parts must be functioning properly. Let X be the amount of time, in months, that each part functions; it is known that X has probability density function

$$p(x) = \begin{cases} cxe^{-x/2} & x > 0, \\ 0 & x \leq 0. \end{cases}$$

(a) Determine the value of c.

(b) Determine the probability that each part functions for no more than six months.

(c) Hence, or otherwise, determine the probability that the whole machine functions for at least six months.

(a) Since $p(x)$ is a probability density function, we have

$$\int_{\mathbb{R}} p(x)\, dx = 1 \quad \Longrightarrow \quad \int_0^\infty cxe^{-x/2}\, dx = 1.$$

We integrate by parts: letting $u = x$ and $dv = e^{-x/2}\, dx$, with $du = dx$ and $v = -2e^{-x/2}$, we have

$$\begin{aligned} \int_0^\infty cxe^{-x/2}\, dx &= c\left[-2xe^{-x/2} + 2\int e^{-x/2}\, dx\right]_0^\infty = c\left[-2xe^{-x/2} - 4e^{-x/2}\right]_0^\infty \\ &= c\left[-2\left(\lim_{x\to\infty}\left(\frac{x}{e^{x/2}} - 2e^{-x/2}\right)\right) - (0-4)\right]. \end{aligned}$$

Note that since $e^{x/2} \to \infty$ as $x \to \infty$, we can use l'Hôpital's rule to evaluate the first limit:

$$\lim_{x\to\infty} \frac{x}{e^{x/2}} = \lim_{x\to\infty} \frac{1}{e^{x/2}/2} = 0.$$

Hence, we require that

$$\int_0^\infty cxe^{-x/2}\, dx = 4c = 1 \quad \Longrightarrow \quad c = \frac{1}{4}.$$

(b) We consider the probability that a single part lasts for less than 6 months:

$$\begin{aligned} P(X < 6) &= \int_0^6 \frac{1}{4}xe^{-x/2}\, dx = \frac{1}{4}\left[-2xe^{-x/2} - 4e^{-x/2}\right]_0^6 \\ &= \frac{1}{4}\left[-2\cdot 6\cdot e^{-3} - 4e^{-3} + 4\right] = 1 - 4e^{-3}. \end{aligned}$$

(c) Let Y be the number of parts that last for at least six months. Since the components function independently of one another, we have $Y \sim \text{Binom}\left(5, 4e^{-3}\right)$. The probability that the system works is thus

$$\begin{aligned} P(Y \geq 3) &= P(Y = 3) + P(Y = 4) + P(Y = 5) \\ &= \binom{5}{3}\left(4e^{-3}\right)^3\left(1 - 4e^{-3}\right)^2 + \binom{5}{4}\left(4e^{-3}\right)^4\left(1 - 4e^{-3}\right) + \binom{5}{5}\left(4e^{-3}\right)^5 \\ &\approx 0.05726796\ldots. \end{aligned}$$